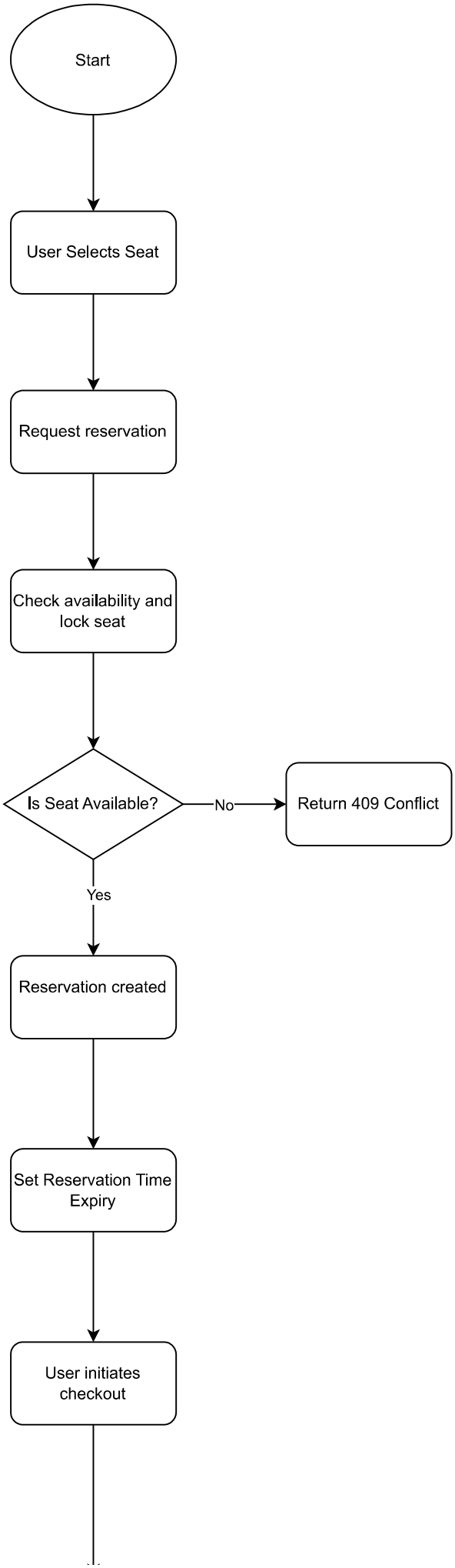
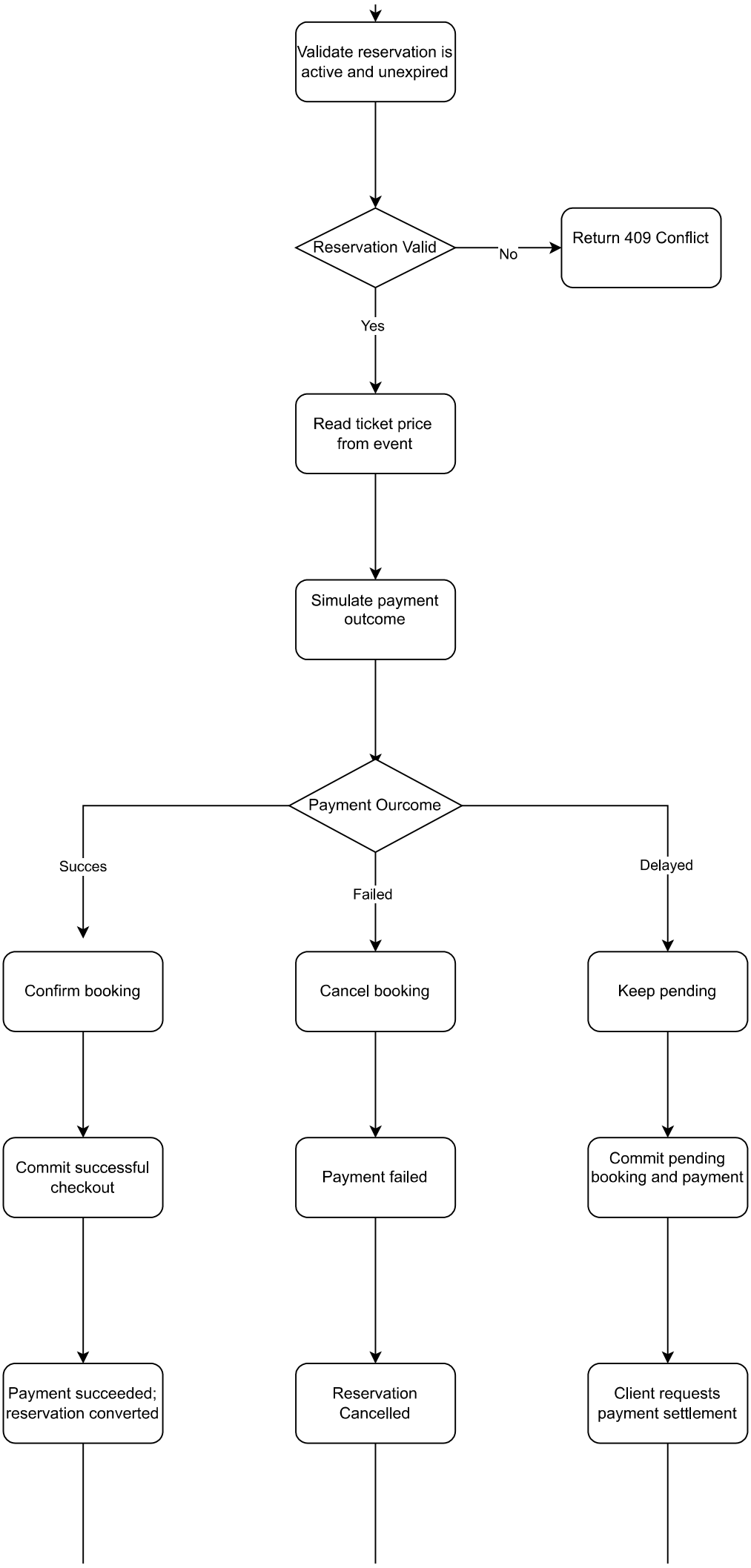
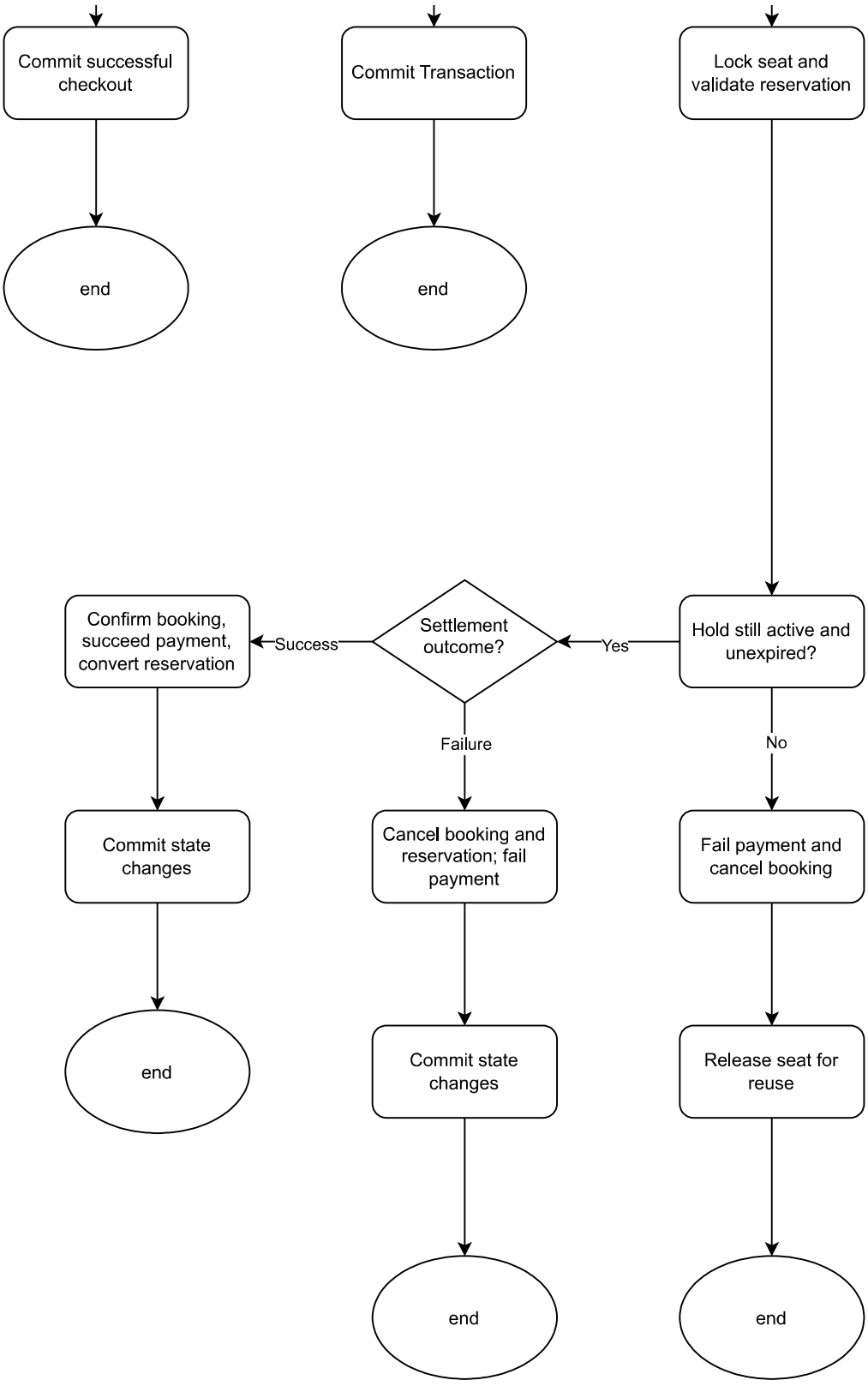

Start
User Selects Seat
Request reservation
Check availability and lock seat
Is Seat Available?
No
Return 409 Conflict
Yes
Reservation created
Set Reservation Time Expiry
User initiates checkout


Validate reservation is active and unexpired
Reservation Valid
No
Return 409 Conflict
Yes
Read ticket price from event
Simulate payment outcome
Payment Ourcome
Succes
Failed
Delayed
Confirm booking
Cancel booking
Keep pending
Commit successful checkout
Payment failed
Commit pending booking and payment
Payment succeeded; reservation converted
Reservation Cancelled
Client requests payment settlement


Commit successful checkout
end
Commit Transaction
end
Lock seat and validate reservation
Hold still active and unexpired?
Yes
Settlement outcome?
Success
Confirm booking, succeed payment, convert reservation
Commit state changes
end
Failure
Cancel booking and reservation; fail payment
Commit state changes
end
No
Fail payment and cancel booking
Release seat for reuse
end